

ARRAYS

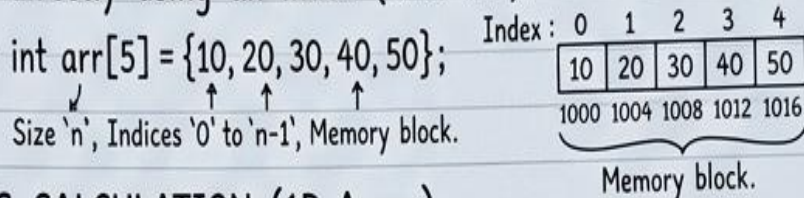
Date: 12/10/23 | Subject: DSA (Data Structures)

✓ Revision 01

① INTRODUCTION & DEFINITION ↩

An array is a collection of homogeneous elements (same data type) stored in contiguous memory locations. Elements can be accessed directly using an index (0 to n-1).

Example: `int arr[5] = {10, 20, 30, 40, 50};`



② ADDRESS CALCULATION (1D Array)

To find memory address of `arr[i]`:

$$\text{Addr}(\text{arr}[i]) = \text{BaseAddr} + (i * \text{size_of_element})$$

← Imp formula for GATE!

Example: Find addr of `arr[3]` if Base=1000, size=4B.
 $1000 + (3 * 4) = 1012$

③ ADVANTAGES (Pros)

1. Direct/Constant Time Access ($O(1)$) via index. ← Imp!
2. Memory efficiency (contiguous storage). ← Recall!
3. Cache friendliness (due to spatial locality).

④ DISADVANTAGES (Cons)

1. Static size (declared at compile time, hard to resize).
2. Inserting/Deleting an element is expensive ($O(n)$) due to shifting elements.
3. Memory must be contiguous (can cause fragmentation).

⑤ IMPORTANT GATE POINTS ★

- Complexity: Access $O(1)$, Insertion/Deletion $O(n)$, Search $O(n)$ or $O(\log n)$ (Sorted) ← Recall!
- Multidimensional Arrays: 'row-major' and 'col-major' order (Formulas needed!)
- '`arr[i]`' is same as '`*(arr + i)`'
- `Sizeof(arr)` gives total size, `Sizeof(arr[0])` gives single element size.

⑥ COMMON QUESTIONS ?

Gate 2019 Q

- Q1: Diff between Static vs Dynamic Arrays? (Dynamic: resizing `std::vector`, `ArrayList`)
- Q2: Matrix multiplication complexity? $O(N^3)$
- Q3: How to check if Array is sorted? (Check `arr[i] <= arr[i+1]` for all i)